Requirements Specifications

Pet Boarding Management System

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1. Introduction

Purpose

This document will define all functional and non-functional requirements for the Pet Boarding Management System enhancement. It will be developed as a full featured desktop application for managing pets, users, and boarding stays.

System Background

The Pet Boarding Management System is a Java Swing desktop application backed by SQLite. It will support:

- Secure user authentication
- Pet intake and management
- Search and sort capabilities
- Role-based access control
- GUI navigation

Objectives and Goals

The completed system should allow a pet boarding facility to securely manage users, register and track boarded pets, search and sort records through an intuitive graphical interface. It must support secure authentication, role-based access, accurate data storage, and efficient retrieval of information.

To achieve this, the system design must include the following:

- Implement a secure login and authentication process using hashed and salted passwords.
- Develop a modular GUI that provides consistent navigation across all system features.
- Create full CRUD functionality for pets and users, including validation and database persistence.

- Implement search and sorting algorithms to filter and organize pet records efficiently.
- Enforce role-based access control to restrict or permit actions based on user type.
- Ensure database reliability through structured DAO components and error handling.

System Description

The system is a standalone desktop application. It replaces a console-based prototype and introduces a layered architecture:

- UI Layer – Swing frames and panels
- Service Layer – Business logic
- DAO Layer – Database operations
- Database Layer – SQLite

System Scope

Core System Functions

1. Login & Authentication
2. Main GUI Navigation
3. Pet Intake & Management
4. Search & Sort
5. User & Role Management
6. Potential Future Enhancements Time Permitting (billing, reports, vaccination tracking, etc.)

User Roles and Permissions

User Type	Description	Permissions
Admin	Full system access	All features
Staff	Standard employee	Pet intake, search, billing
Read-Only	View-only access	No modifications

Constraints

- Must run on Windows/Mac with Java 17+
- Must use SQLite
- Must use Swing for GUI
- Passwords must be stored securely

Assumptions

- Only local database access is required
- Users have basic computer skills
- Facility operates with a single workstation

Functional Requirements

Login & GUI Requirements

1. **Login Screen**
    - The system shall display a login screen at startup.
    - The system shall provide username and password fields.
    - The system shall validate that both fields are not empty.
    - The system shall authenticate credentials using AuthService.
    - The system shall display an error message for invalid credentials.
    - The system shall open the Main Window upon successful login.
    - The system shall allow pressing Enter to trigger login.
2. **Authentication**
    - Passwords shall be stored as salted, hashed values.
    - The system shall retrieve user records via UserDAO.
    - The system shall compare hashed input passwords with stored hashes.
    - The system shall create a default admin user if none exist.

3. **Main GUI Shell**
    - The system shall display a menu bar with navigation options.
    - The system shall show the logged-in username and role.
    - The system shall allow logout and exit.

Pet Intake & Management Requirements

1. **Pet Intake**
    - The system shall allow adding new pets.
    - The system shall validate required fields (name, type, owner).
    - The system shall store pet information in the database.
2. **Pet Editing**
    - The system shall allow editing existing pet records.
    - The system shall validate updated fields.
3. **Pet Removal**
    - The system shall allow deleting pet records.
    - The system shall require confirmation before deletion.

Search & Sort Requirements

1. **Search**
    - The system shall allow searching pets by:
        - Name
        - Owner
        - Type
    - The system shall display results in a table.

2. **Sorting**

- The system shall allow sorting by:
  - Name
  - Check-in date
- Sorting shall use efficient algorithms (merge sort).

User & Role Management Requirements

1. **User Management**

- Admins shall create new users.
- Admins shall assign roles.
- Admins shall deactivate users.

2. **Role Enforcement**

- Staff users shall not access admin screens.
- Read-only users shall not modify data.

Non-Functional Requirements

Performance

- Login must complete within 2 seconds.
- Database queries must return within 1 second.

Security

- Passwords must use PBKDF2WithHmacSHA256.
- Passwords must include a unique salt.

Usability

- GUI must be intuitive and accessible.
- Error messages must be clear.

Reliability

- The system must handle database failures gracefully.
- The system must not crash due to invalid input.

Maintainability

- Code must follow layered architecture.
- UI, service, and DAO layers must be decoupled.